Problem Chosen	2020 ICM Summary Sheet	Team Control Number
E		2016845

Summary

Since the 1950s, the plastic industry has developed rapidly and has entered all corners of human life. Its positive role is undoubted but it also brings negative effects. Most disposable plastics are non-degradable, which damages both land and sea, disrupts the balance of the earth's ecosystem and threatens human health. This paper analyzes the problems from a wide range of perspectives and proposes achievable solutions.

For task 1, the environmental capacity of disposable plastic waste is defined as the sum of harmless incineration, recycling, and natural self-purification. This paper builds the MIRP model. In order to quantify the above three variables, we put forward core indicators. We use the Multivariate Nonlinear Regression model to find the formulas of harmless incineration, recycling and nature self-purification' regression and their indicators. Finally, the three regression formulas were added together to obtain the environmental capacity formula. By using particle swarm optimization, the maximum environmental capacity is obtained.

For task 2, First, to reach an environmentally safe level is to make the difference between the original disposable plastic demand and the disposable plastic waste less than the environmental capacity obtained in question one. Second, analyze three impacts during the reduction process, the source and use of disposable plastic waste, residents' satisfaction, environmental policies, and regional differences. This paper takes disposable plastic waste reduction as the dependent variable and their indicators, policies, and measures as independent variables. Analyze their relation by regression. The Simulated Annealing algorithm was used to find the optimal solution.

For task 3, set a minimum level for the global disposable plastic waste based on the environment capacity obtained in task 1. In order to achieve the goal, some policies and measures are formulated. This paper analyzes the impact on the lives of residents, the ecological environment, and the development of the plastics industry. The results show that under the implementation of plastic-restriction policies and measures, residents' satisfaction has shown a trend of first decline and then rise, the environment has shown an upward trend, and the development of the plastic industry has shown a downward trend.

For task 4, the solution of global disposable plastic waste needs to consider how different countries or regions should fairly shoulder their responsibilities. We choose six positive and negative indicators to represent the realistic situation on environment, economy, and policy. We use the Range Transform method to process the indicators. Next, the coefficients of indicators are determined by the Entropy Weight method. Then, using the historical data and coefficient of each indicator to obtain the score matrix of each country. Normalize the score matrix and times score matrix with the total environmental capacity to obtain the quota for plastic usage for each country or region. Inspired by carbon trading rule, the plastic usage quotas are open to trade between the quota surplus side and scarce side. It considers both fairness and environmental protection.

Keywords: disposable plastic waste; environmental capacity; MIRP model

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1. Introduction

1.1 Background

Since the 1950s, the plastic industry has developed rapidly in the world, promoting the development of society; in the past decade, the rise of take out industry has provided convenience for human life, and the positive role of plastic is undoubted. But the development of plastic industry also has its disadvantages. Most of the plastics are not easy to degrade, which makes the land and ocean affected, destroys the balance of the earth's ecosystem, and damages human health. Therefore, the problem of plastic waste pollution needs to be solved. In this context, this paper analyzes the problem of plastic pollution from multiple perspectives, and puts forward corresponding solutions.

1.2 Restatement of the Problem

In order to solve the problem of plastic pollution, the following tasks are to be accomplished:

- A model to estimate the maximum levels of single-use or disposable plastic product waste that can safely be mitigated without further environmental damage is needed. And it needs to be combined the source of this waste, the extent of the current waste problem, and the availability of resources to process the waste. We should fully consider two aspects: human processing ability and nature's own purification ability. (Refer to Task 1)
- It is necessary to discuss how much plastic use can be reduced through some policies and measures, and analyze the relationship between the amount of plastic waste reduction and the policies and measures implemented; at the same time, it is also necessary to consider the impact of regional factors on the effectiveness of policies and measures.(Refer to Task 2)
- Setting a goal for the global minimum level of achievable plastic product use. In order to achieve this goal, the impact on lifestyle, ecological environment and plastic industry development will be analyzed. (Refer to Task 3)
- Plastic waste pollution is a global problem, it needs to determine how different countries should bear the corresponding responsibilities fairly. (Refer to Task 4)
- Write a two-page memo to the ICM to introduce the time point to achieve the goal in Task 3, and analyze what factors will affect the realization of the target. (Refer to Task 5)

2. Assumptions

- We assume that all the data collected are true
- The main productive forces and social structure won’t change in recent years.

3. Notations

The following table lists the symbols used in this paper and their meanings

Table 1 Notations of the model

Symbols	Definition
M	the maximum levels of single-use or disposable plastic product waste
I	Incineration
R	Recycled
P	Nature's ability of self purification
GDP	Gross domestic product
N	Number of incineration harmless treatment plants
PEAL	Public environment awareness level
TC	Total investment in environmental pollution control
RLI	Reverse logistics input
E	Effectiveness of Ecological Civilization
T_A	Annual average temperature
R_A	Average annual rainfall
P_R	Plastic waste reduction
OPD	Original plastic demand
APC	Alternative disposable plastic product consumption
HI	Happiness Index
BPP	Banned plastic production
IRP	Imports of recycled plastics
DPRR	Disposable plastic replacement ratio
EB3P	Efficiency of Banning Plastic Production Policy
EI	the Environmental Impacts
IPID	the Influence of Plastic Industry Development
QLR	the Quality of Life of Residents
PD	the Density of Population
EPI	Environmental performance index
GPI	Global Peace Index
IP	Industrial production
C	Final consumption expenditure per capita

4. General framework

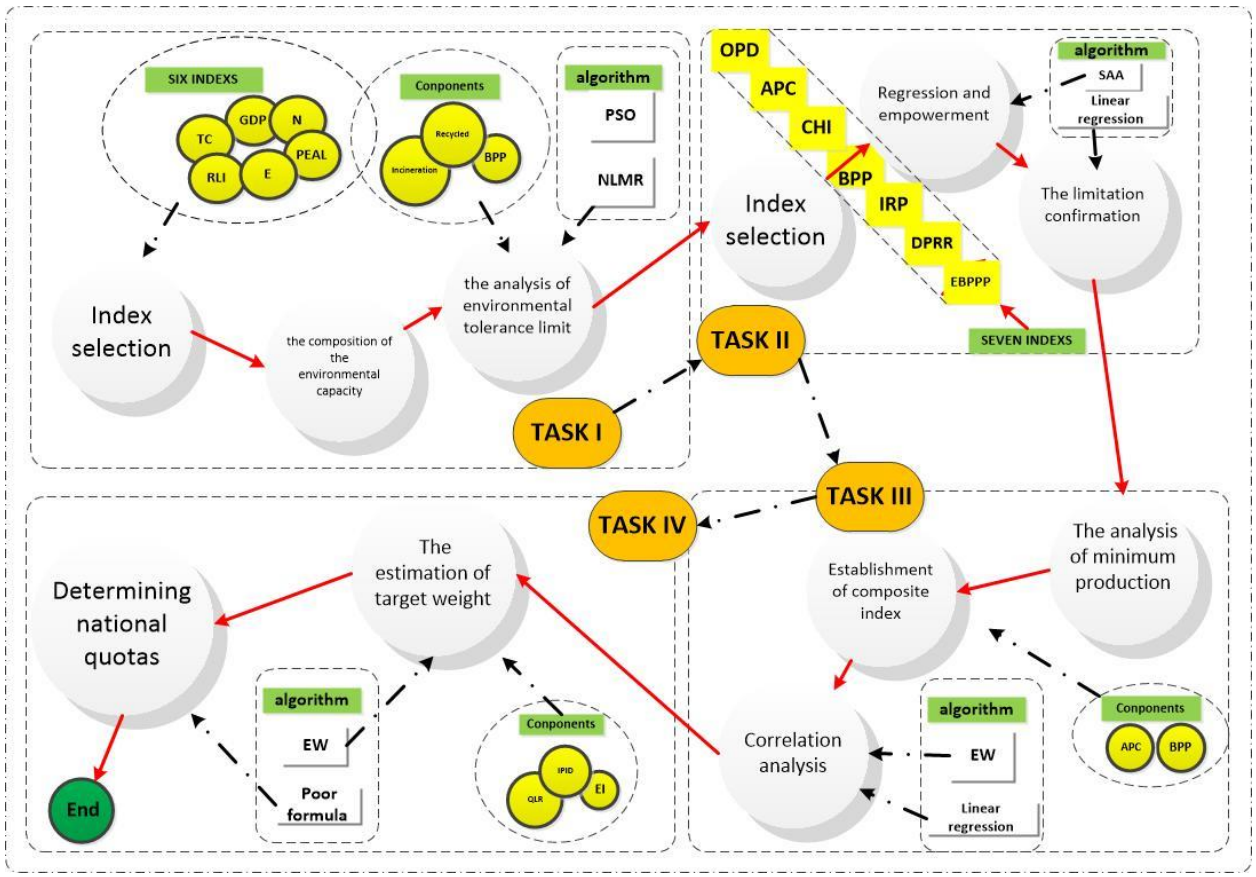

Figure 1 General framework

5. MIRP model based on multiple nonlinear regression model and particle swarm optimization algorithm

For problem 1, the maximum levels of single-use or disposable plastic product waste can be defined as the maximum levels of the total waste disposal based on the existing manual waste disposal ability and nature's ability of self-purification. If the total amount of disposable plastic exceeds this maximum level, it brings further environmental damage. Otherwise the waste can safely be mitigated[1].

This paper sets up a model to estimate the maximum levels of single-use or disposable plastic product waste that can safely be mitigated without further environmental damage. It considers many factors, sources of disposable plastic waste, capacity of manually processing the disposable plastic waste, availability of resources to process the waste, and current status of environmental issues, etc.

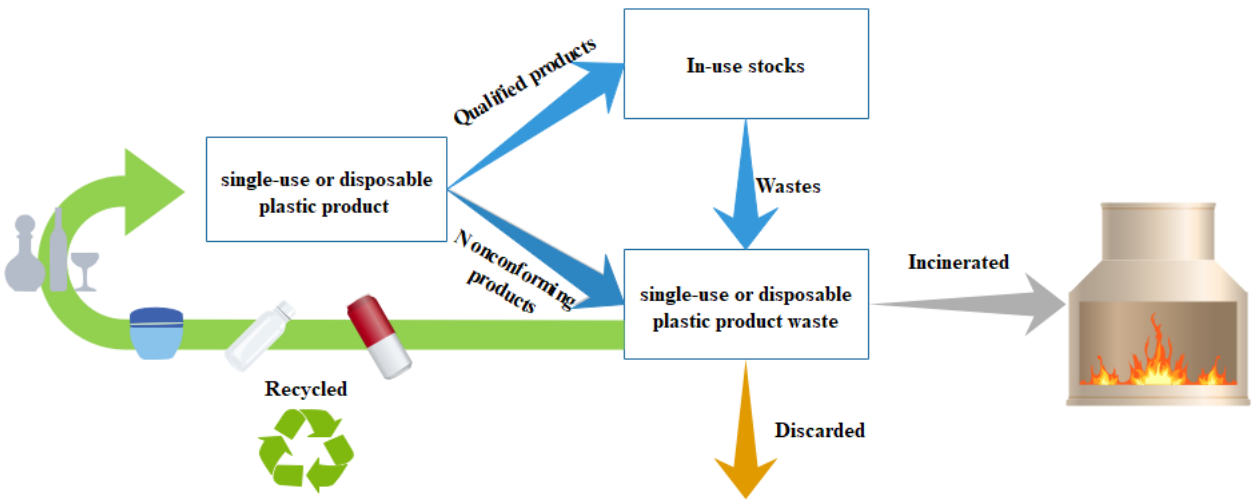

Figure 2 Life cycle of disposable plastics

Firstly, the maximum levels of single-use of disposable plastic product are divided into three parts, harmless incineration of disposable plastic product waste, recycling of disposable plastic product waste and nature's ability of self-purification [2]. Next, find out the influencing factors of each part and establish their equations respectively based on Multivariate Nonlinear Regression Model. Through multiple experiments, it is found that the effect of multiple nonlinear regression in the three analytical solutions is at the highest level. Then, the sum of three analytical formulas represents the maximum levels of the total waste disposal made by human and environment, which is also the formula of maximum safe levels of disposable plastic waste [3]. Finally, list constraints on these impact factors based on the considerations of residents' needs, economic status, ecological environment construction, etc. [4]. Therefore, we solve the problem of finding the maximum levels of single-use or disposable plastic product waste(M)by Particle swarm optimization。

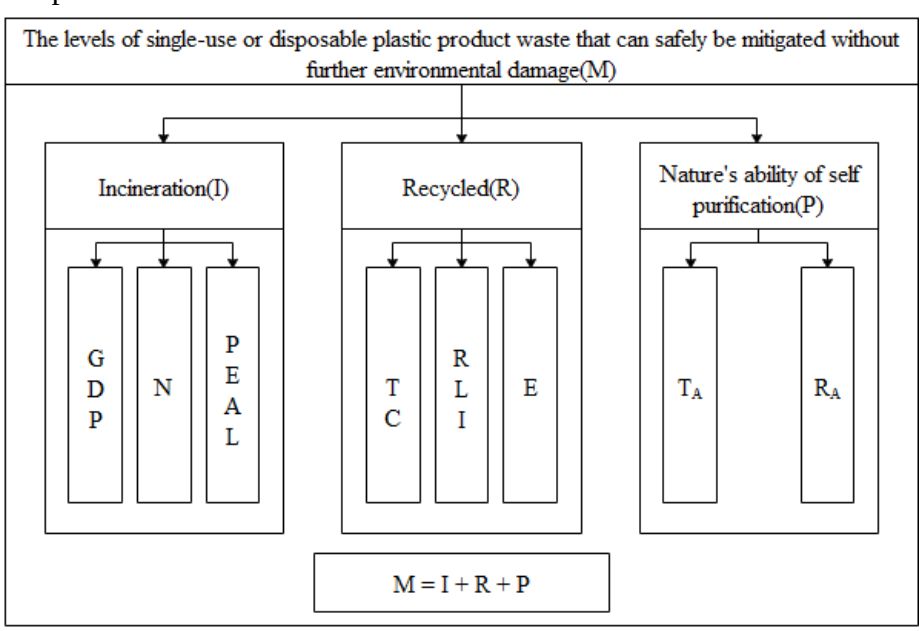

Figure 3 The composition of environmental capacity

5.1 Introduction to model and algorithm

5.1.1 Multivariate nonlinear regression model

Multiple nonlinear regression analysis refers to a nonlinear regression model that includes more than two variables. The traditional approach is to transform it into a standard linear form of multiple regression models. Some non-linear regression models can obtain linearized expressions through proper mathematical transformation, while other non-linear regression models cannot. The nonlinear regression model of the former case is generally called **Intrinsic Linear Regression**, while the latter is called **Intrinsic Non-linear Regression**. This paper uses **Intrinsic Non-Linear Regression**, which has better fitting effect.

5.1.2 Particle Swarm Optimization

Particle Swarm Optimization, also known as Bird Population Foraging Algorithms, abbreviated as PSO, is one of the Evolutionary Algorithm – EA developed by J. Kennedy and R.

Similar to Simulated Annealing, it also starts from random solutions, finds the optimal solution by iteration, and evaluates the solution by adaptability level. It is simpler than the Genetic Algorithm because it does not have the "crossover" and "mutation" operation of the genetic algorithm. It looks for global optimal value by finding the current optimal value among all searched items. PSO is popular among scholars because of its low processing difficulty, high precision, and fast convergence. Particle swarm optimization algorithm flow is shown below.

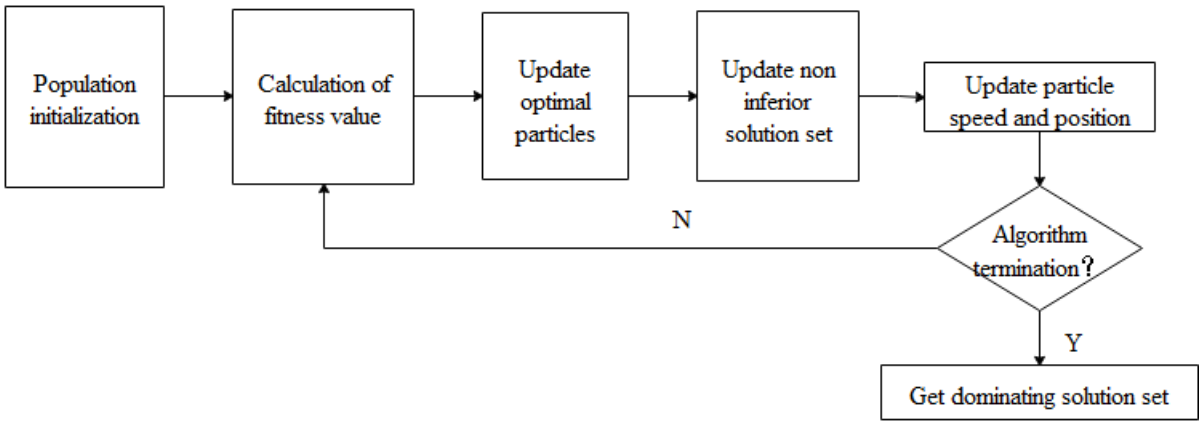

Figure 4 Particle swarm optimization algorithm flow

5.2 Multivariate nonlinear regression model and application of particle swarm optimization

Firstly, identifying the influencing factors in three parts, harmless incineration of disposable plastic product waste, recycling of disposable plastic product waste and nature's ability of self-purification. By consulting the literature, harmless incineration of disposable plastic product waste is closely related to the number of incineration plants, the level of public environmental awareness and GDP; the recycling of disposable plastic product waste is closely related to the total investment in environmental pollution treatment, reverse logistics investment, and ecological civilization construction; nature's ability of self-purification is closely related to temperature, humidity, and microbial density. Then, find the historical data of each indicator and further analyze the relationship between the three parts and their respective influencing factors. This article uses China as an example. The index data are calculated based on Chinese calendar years.

$$M = I + R + P \tag{1}$$

For part 1, based on the data from 2009 to 2018 of China 's GDP, the number of incineration plants, and the existing quantitative research results of public environmental awareness, the public environment awareness level formula is as followed. Bringing $x = 2009, 2010, \ldots, 2018$ into Equation 2 can get public environmental awareness for the corresponding year.

$$PEAL = 2E - 16e^{-0.0198x} \tag{2}$$

Based on the data from 2009 to 2018 of China 's GDP, the number of incineration plants(N), the level of public environmental awareness($PEAL$) and harmless incineration of disposable plastic product waste(I), we seek to devise two models, Multiple Linear Regression and Multiple Nonlinear Regression, for exploring the relationship of these indicators. The results show that the fitting effect of multivariate nonlinear regression is better with $SSE = 1.41(E - 25)$, which is shown in formula (3). The fitted image is shown in Figure 5.

$$\begin{aligned} I = {} & 3.216 + 15.51 \times N - 161 \times PEAL + 1.387(E + 04) \times N \times PEAL \\ & - 6097 \times PEAL^2 + 8.696(E + 04) \times N^3 - 2.319(E + 05) \times N^2 \times PEAL \\ & + 1.816(E + 05) \times N \times PEAL^2 - 4.425(E + 04) \times PEAL^3 - 5874 \times N^2 \end{aligned} \tag{3}$$

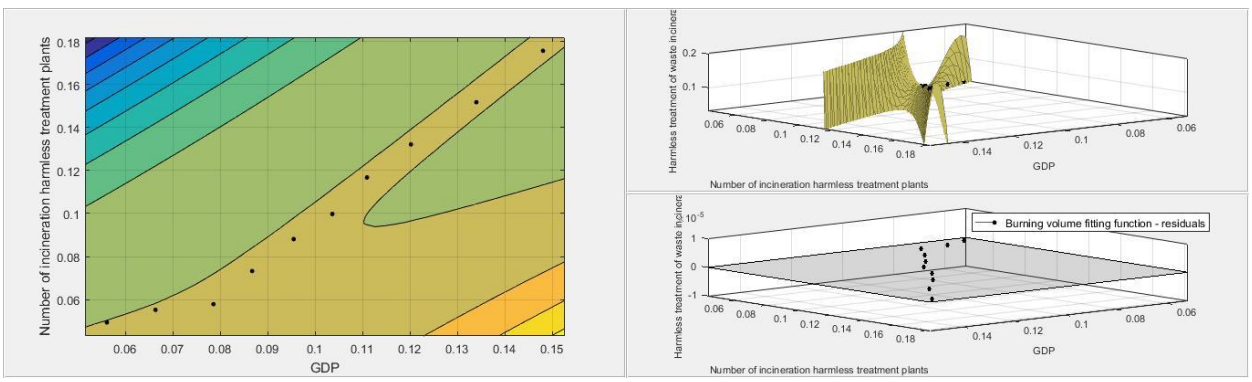

Figure 5 The fitting figure for I

Based on the data from 2009 to 2018 of total investment in environmental pollution control(TC) , reverse logistics input (RLI) and the existing research result of effectiveness of ecological civilization(E), we further compute the quantitative result of ecological civilization construction from 2009 to 2018 in formula(4).

$$E = \sum_{i=1}^{n} Y_i \times \sum_{i=1}^{n} \sum_{j=1}^{m} W_{ij} P_{ij} \tag{4}$$

Based on the data from 2009 to 2018 of recycling of disposable plastic product waste(R) , total investment in environmental pollution control(TC) , reverse logistics input (RLI) and the existing research result of effectiveness of ecological civilization(E), we build many regression models for R and the results show that Multiple Nonlinear Regression is the best method with $SSE = 7.097(E - 28)$. The fitting figure and the fitting result are as follows.

$$\begin{aligned} R = {} & 1.923 - 101.7 \times TC + 45.92 \times RLI + 1598 \times TC^2 - 8.787 \times RLI^3 + 203.8 \times RLI^2 \\ & - 7343 \times TC^3 + 7199 \times TC^2 \times RLI - 1677 \times TC \times RLI^2 - 1244 \times TC \times RLI \end{aligned} \tag{5}$$

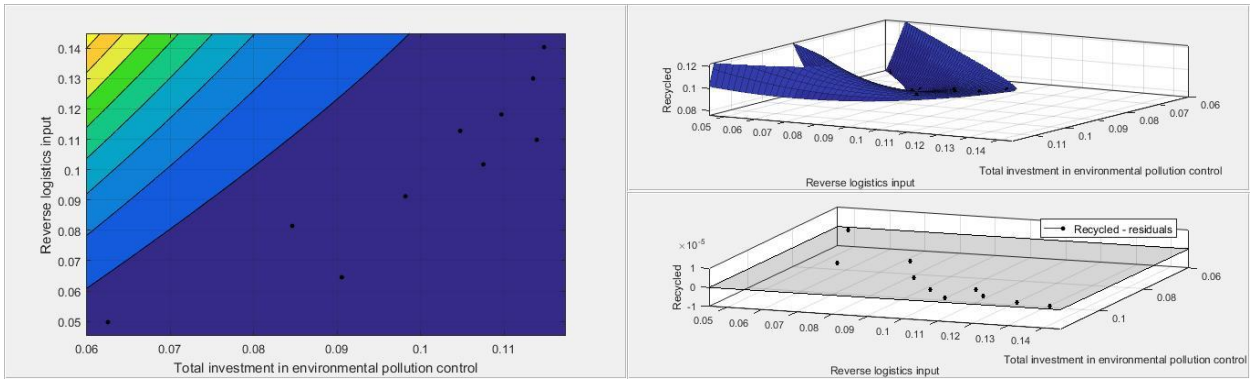

Figure 6 The fitting figure for R

Based on the data from 2009 to 2018 of nature's ability of self-purification(P), annual average temperature(T_A) and average annual(R_A,) we build many regression models for P and the results show that Multiple Nonlinear Regression is the best method with $SSE = 0.005511$. The fitting figure and the fitting result are as follows.

$$\begin{aligned} P = & -6963 + 2.43(E+05) \times T_A + 4.171(E+04) \times R_A - 3.065(E+06) \times T_A^{\ 2} \\ & -1.294(E+06) T_A \times R_A + 1.628(E+07) \times T_A^{\ 3} + 1.336(E+07) \times T_A^{\ 2} R_A \\ & -2.963(E+07) \times T_A^{\ 4} - 4.583(E+07) \times T_A^{\ 3} R_A \end{aligned} \tag{6}$$

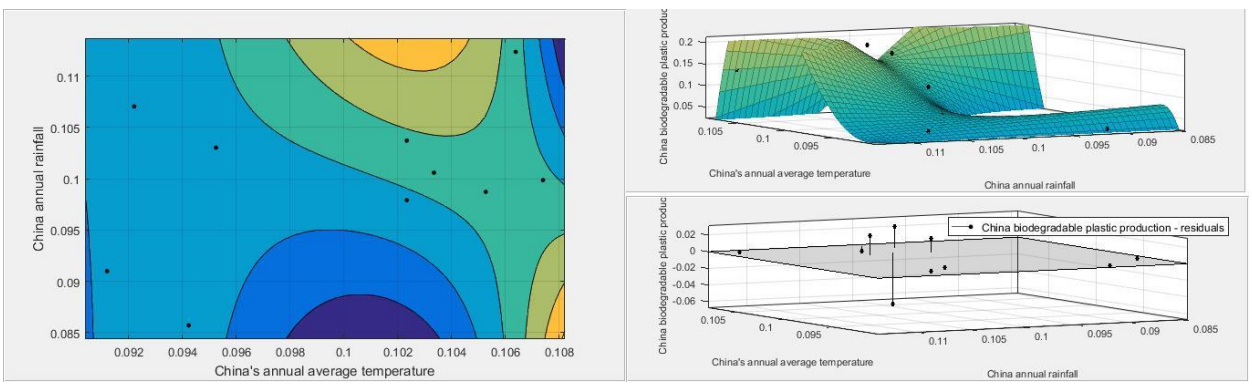

Figure 7 The fitting figure for P

Finally, list constraints of three analytical formula, the geographical constraints of the number of incineration treatment plants, the constraints of residents' living demands, and the economic development limitation of total investment in environmental pollution control. Then use Particle Swarm algorithm to solve the problem. The optimization result shows that the maximum of Incineration is $I_{MAX} = 15432$ (kiloton) if the number of incineration harmless treatment plants $N = 212$ and public environment awareness level $PEAL = 0.7861$. The iteration process is demonstrated by figure 8.

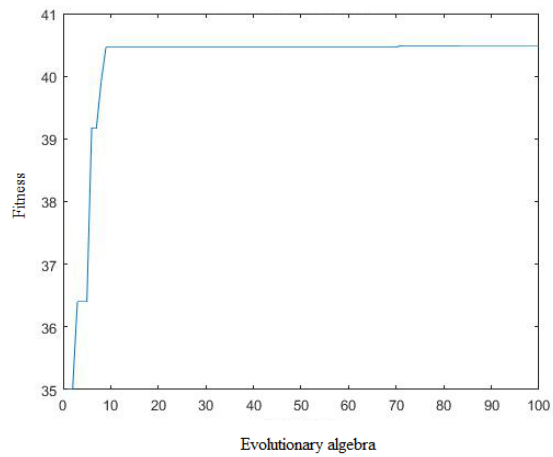

Figure 8 Optimization process for I

In the same sort of way, the maximum of Recycle $R_{MAX} = 15623$ （kiloton）is obtained if $TC = 98725$ （myriads）and $RLI = 124069$（myriads）. The optimization process is shown in figure 9.

And by the same work way, the maximum of Nature's ability of self-purification $P_{MAX} = 1800$ (kiloton) is calculated if $T_A = 10.31$ (Celsius) and $R_A = 662.5$ (millimeter). The optimization process is shown in figure 10.

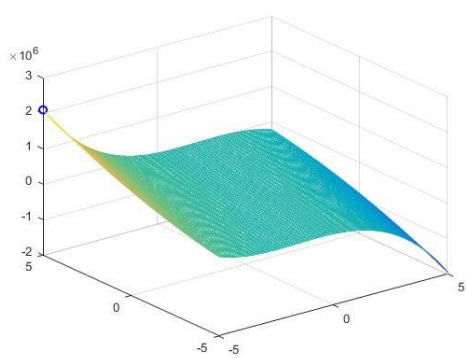

Figure 9 Optimization process for R

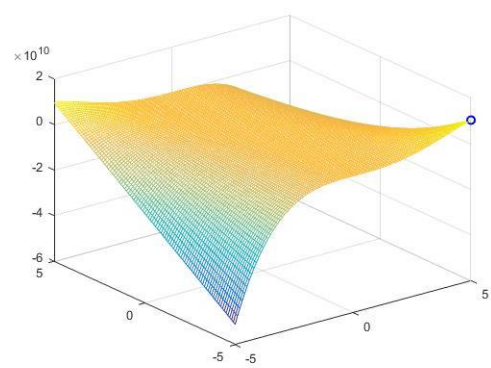

Figure 10 Optimization process for P

Overall, M, the maximum levels of disposable plastic product waste that can safely be mitigated without further environmental damage, is calculated as $M = 32855$ (kiloton). We can also say that the maximum environmental capacity of disposable plastic waste in China is 32855(kiloton) based on our first analysis. When the waste exceeds this level, the environment would be harmed.

6. The discussion of plastic waste reduction

6.1 Basic analysis

To solve question 2, we need to analyze two basic aspects. The first one is to what extent plastic waste can be reduced to reach an environmentally safe level. Another is the impact of source and use of disposable plastic waste, residents' satisfaction, environmental policies (For

example, the ban of some non-degradable plastic products production, plastic recycling policies, encouraging the application of biodegradable plastics, the application of alternatives, etc.), geographical differences. Therefore, this paper takes the reduction of plastic waste as a dependent variable, policy and regional-specific constraints as the influencing factor, analyzes the relationship between plastic waste reduction and policy and regional-specific constraints by regression model. The reduction of plastic waste which makes environment safe can be calculated by Simulated Annealing model. At the same time, in order to ensure the quality of ecological environment, it is also required that the difference between the original plastic demand (OPD) and the plastic waste reduction (P_R) is less than the maximum levels of single-use or disposable plastic product waste (M) in the first question.

6.2 Optimization model based on sigmoid function and Simulated Annealing algorithm

6.2.1 The establishment of optimization model

We analyze that the reduction of plastic waste (P_R) is related to policies. There are two main kinds of plastic policies, banning the production of some non-degradable plastic and encouraging the use of disposable plastics alternatives. The two policies' efficiency is expressed by two indexes, the alternative disposable plastic product consumption (APC) and banned plastic production (BPP) respectively. However, the sum of APC and BPP is not equal to the reduction of plastic waste (P_R). So, we further introduce two adjustment coefficients, disposable plastic replacement ratio ($DPRR$) and efficiency of banning plastic production policy ($EB3P$). The formula is shown below.

$$P_R = APC \times DPRR + BPP \times EB3P \tag{4}$$

6.2.2 The parameters determination model based on sigmoid function

This paper introduces Logical Regression Model to determine the value of the two coefficients. By consulting the literature, we find that three indicators, public environment awareness level($PEAL$), happiness index (HI) and GDP are related to one coefficient ($DPRR$ - disposable plastic replacement ratio) . We also notice that three indicators, total investment in environmental pollution control (TC), banned plastic production(BPP) and the imports of recycled plastics (IRP) are tightly related to another coefficient ($EB3P$ - efficiency of Banning Plastic Production Policy). We collect data of six indicators in China between 2009 and 2018. The following matrix express the coefficients. Each column of the matrix represents the normalized value of one indicator in ten years since the data of different indicators are in different units.

$$T_{DPRR} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ \vdots & \vdots & \vdots \\ x_{10,1} & x_{10,2} & x_{10,3} \end{pmatrix} \qquad T_{EB3P} = \begin{pmatrix} y_{11} & y_{12} & y_{13} \\ y_{21} & y_{22} & y_{23} \\ \vdots & \vdots & \vdots \\ y_{10,1} & y_{10,2} & y_{10,3} \end{pmatrix}$$

Since the data of different indicators are in different units, each column of the matrix represents the normalized value of one indicator in ten year. The normalization process is as follows.

$$GDP' = \frac{\sum_{i=1}^{n=10} x_{i1}}{10 \times \sum_{i=1}^{n=10} \sum_{j=1}^{m=3} x_{ij}} \qquad TC' = \frac{\sum_{i=1}^{n=10} y_{i1}}{10 \times \sum_{i=1}^{n=10} \sum_{j=1}^{m=3} y_{ij}}$$

$$PEAL' = \frac{\sum_{i=1}^{n=10} x_{i2}}{10 \times \sum_{i=1}^{n=10} \sum_{j=1}^{m=3} x_{ij}} \qquad BPP' = \frac{\sum_{i=1}^{n=10} y_{i2}}{10 \times \sum_{i=1}^{n=10} \sum_{j=1}^{m=3} y_{ij}}$$

$$HI' = \frac{\sum_{i=1}^{n=10} x_{i3}}{10 \times \sum_{i=1}^{n=10} \sum_{j=1}^{m=3} x_{ij}} \qquad IRP' = \frac{\sum_{i=1}^{n=10} y_{i3}}{10 \times \sum_{i=1}^{n=10} \sum_{j=1}^{m=3} y_{ij}}$$

By consulting the literature about the relation degree, we estimate the weight of each indicators by weighted average method. The formula are show below.

$$DPRR' = 0.4 \times GDP' + 0.4 \times PEAL' + 0.2 \times HI' \tag{5}$$

$$EB3P' = 0.4 \times TC' + 0.3 \times BPP' + 0.3 \times IRP' \tag{6}$$

Since the value of coefficients may greater than 1, we need to map the coefficients to 0-1 intervals. In information science, sigmoid function, a common S-type function in biology which graph called S-type growth curve, is often used as activation functions of neural network because of its and its inverse function characteristics of monotonic increasing. In this case, we can eliminate the possibility of exceeding the 0-1 range by bringing $DPRR'$ and $EB3P'$ into the sigmoid function and obtain $DPRR$ and $EB3P$.

$$S(x) = \frac{1}{1 + e^{-x}} \tag{7}$$

$$DPRR = \frac{1}{1 + e^{-DERR'}} \tag{8}$$

$$EB3P = \frac{1}{1 + e^{-EB3P'}} \tag{9}$$

Considering the huge geographical and cultural differences in China, we calculate the disposable plastic replacement ratio and the efficiency of Banning Plastic Production Policy of city and countryside respectively. The results show that $DPRR = 0.6293$, $EB3P = 0.5900$ in the countryside and $DPRR = 0.6167$, $EB3P = 0.6138$ in the city. Obviously, the efficiency of plastic policies varies from regions. The urban's $DPRR$ is less than the countryside's because fast urban life space lead to the high demand of take-out food which are often packaged by

disposable plastic bags. While the urban's $EB3P$ is greater than the countryside's, the contribution for this may related to the well-developed urban communication.

6.2.3 Regression analysis

Now the coefficients are settled, this paper puts forward formulas of alternative

disposable plastic product consumption (APC) and banned plastic production (BPP) of city and countryside respectively. We seek to use regression model to analyze the trends of APC and BPP with years. The results show that the univariate nonlinear regression is the best regression method.

In the city,

$$APC = -76 \times z^2 + 400 \times z + 4300; \text{where } z = (t - 2000) / 3 \tag{10}$$

$$BPP = 6.4z^2 + 140 \times z + 1000; \text{where } z = (t - 2000) / 3 \tag{11}$$

In the countryside,

$$APC = -58 \times z^2 + 210 \times z + 3100; \text{where } z = (t - 2000) / 3 \tag{12}$$

$$BPP = 0.36 \times t^2 - 1400 \times t + 1360589 \tag{13}$$

6.2.4 Optimal decision model based on Simulated Annealing algorithm

Bring the coefficients and formulas into the equation below:

$$P_R = APC \times DPRR + BPP \times EB3P$$

If the result shows that $OPD - P_R \leq M$. That means our model successfully guarantee the environment safe level.

Then we use Simulated Annealing algorithm to find the solution and the optimal steps are as follows.

Step1: set the initial temperature $T=T_0$ and randomly choose a initial solution x_0. Calculate the value of objective function $E(x_0)$.

Step2: cooling the temperature T_0 to T_i.

Step3: randomly transform the current solution and obtain a new solution x_j. Calculate the corresponding value of objective function $E(x_j)$. Set $\Delta E=E(x_j)-E(x_i)$.

Step4: **If** $\Delta E<0$, x_j is taken and replaces the last solution x_j. If $\Delta E>0$, x_j is taken with probability of $e^{-\Delta E/T_i}$, where T_i is the current temperature.

Step5: Under temperature T_i, repeat the random transformation process(Step 3) and adoption process(Step 4).

Step6: if temperature reach T_f, suspend the operation. Otherwise go to step 2 and continue the operation.

The whole process is operated by MATLAB.

In the city, if $APC_1 = 8350$ (Kilotons) and $BPP_1 = 7986$ (Kilotons), the reduction of plastic $P_{R1} = 10275$ (Kilotons) is the maximum value. The optimization process is show below.

In the countryside, if $APC_2 = 3870$ (Kilotons) and $BPP_2 = 3840$ (Kilotons), the reduction of plastic $P_{R1} = 4760$ (Kilotons) is the maximum value. The optimization process is show below.

Since China's maximum reduction in plastic waste $P_R = P_{R1} + P_{R2} = 15037$ (Kilotons) and $OPD - P_R \leq M$, our model satisfies the environmentally safe level. The final solution also show that the efficiency and specific constraint of plastic pollution control policies varies from regions.

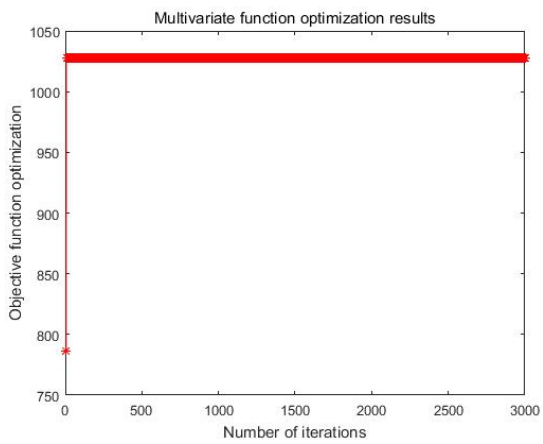

Figure 11 Optimization process for P_{R1}

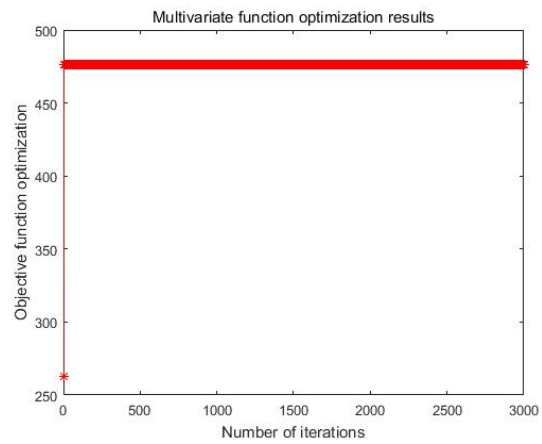

Figure 12 Optimization process for P_{R2}

7. PTGB index analysis based on entropy weight method

7.1 Basic Analysis

Analysis of task 3 is based on the analysis and solutions of task 1 and task2. In task 1, we calculate the maximum levels of single-use or disposable plastic product waste (M) which is also environmental capacity. In task 2, we obtain the plastic waste reduction (P_R) and analyze the relation between P_R and related policies. In task 3, we assume that the global environmental capacity of disposable plastic product can be calculated by the model in task 1. In order to achieve the goal that the global amount of disposable plastic would not exceed the global environmental capacity, some policies and measures should be formulated. Then, we further discuss its impact on the lives of residents, the ecological environment, and the development of the plastic industry.

7.2 Determination of index weights based on Entropy Weight method

In order to research the impact of reducing disposable plastic to environmental safe level, we adopt four indicators in task 1 and task 2, public environment awareness level ($PEAL$), total investment in environmental pollution control (TC), GDP and banned plastic production (BPP). These four indicators are used to measure the extent of impact on the lives of residents, the ecological environment, and the development of the plastic industry.

In order to objectively determine the weight of the above four indicators, this paper introduces Entropy Weight method, thus avoiding the subjectivity of weight.

Step 1: Construct a matrix S, where x_{ij} represents the value of indicator in the year of 2009+i. There are four indicators and data in 10 years (from 2009 to 2018) so, S has 4 columns and 10 rows.

$$S = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ \vdots & \vdots & \vdots & \vdots \\ x_{10,1} & x_{10,2} & x_{10,3} & x_{10,4} \end{pmatrix}$$

Step 2: Calculate the i-th feature weight p_{ij} of the indicator in column j.

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^{10} x_{ij}}$$

Step 3: Calculate the entropy coefficient e_j of j indicator.

$$e_j = -\frac{1}{\ln 10} \sum_{i=1}^{10} p_{ij} \ln p_{ij}$$

Step 4: Calculate the utility indicator u_j.

$$u_j = 1 - e_j$$

Step 5: Calculate the weight h_j of j indicator.

$$h_j = \frac{u_j}{\sum_{j=1}^{4} u_j}$$

Using MATLAB to calculate and get the weight.

$$h = \begin{pmatrix} 0.1284 \\ 0.3579 \\ 0.4420 \\ 0.0716 \end{pmatrix}$$

The weight of four indicators $PEAL$, TC, GDP and BPP are 0.1284, 0.3579, 0.4420 and 0.0716 respectively.

7.3 The construction of PTGB index

Considering the combined effect of four indicators $PEAL$, TC, GDP and BPP, we combined four indicators into one indicator, PTGB index.

Firstly, normalize the matrix S by column and get the matrix S'.

$$S' = \begin{pmatrix} \frac{x_{11}}{\sum_{i=1}^{10} x_{i1}} & \frac{x_{12}}{\sum_{i=1}^{10} x_{i2}} & \frac{x_{13}}{\sum_{i=1}^{10} x_{i3}} & \frac{x_{14}}{\sum_{i=1}^{10} x_{i4}} \\ \frac{x_{21}}{\sum_{i=1}^{10} x_{i1}} & \frac{x_{22}}{\sum_{i=1}^{10} x_{i2}} & \frac{x_{23}}{\sum_{i=1}^{10} x_{i3}} & \frac{x_{24}}{\sum_{i=1}^{10} x_{i4}} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{x_{10,1}}{\sum_{i=1}^{10} x_{i1}} & \frac{x_{10,2}}{\sum_{i=1}^{10} x_{i2}} & \frac{x_{10,3}}{\sum_{i=1}^{10} x_{i3}} & \frac{x_{10,4}}{\sum_{i=1}^{10} x_{i4}} \end{pmatrix}$$

Next, multiply S' and h to get PTGB index matrix. The index reflects the strength of policies of plastic pollution control and measures with positive correlation.

$$PTGB = S' \times h \tag{14}$$

In the following part, this paper analyzes the relation between PTGB index and three affected areas, the lives of residents, the ecological environment, and the development of the plastic industry.

7.4 Univariate Nonlinear Regression model based on PTGB index.

This paper uses 3 indicators, Happiness index, plastic recycling rate and annual growth rate of plastics to represent the impact on the lives of residents, the ecological environment, and the development of the plastic industry respectively. We build a regression model to further study the specific impact.

Firstly, we analyze the relation between Happiness Index and PTGB. After trying different kinds of regression model, Univariate Nonlinear Regression model performs best. The regression equation is shown below.

$$QLR = -0.032 \times z^3 + 0.019 \times z^2 + 0.2 \times z + 5, \quad where \ z = (PTGB - 3.4) / 0.73 \tag{15}$$

As Figure 13 shows, residents are not adapted to restrict the use of plastics in the beginning stage with decreasing trend of Happiness Index. With the promotion of plastic restriction policies and measures, residents feel the benefits brought by plastic restriction with increasing trend of Happiness Index.

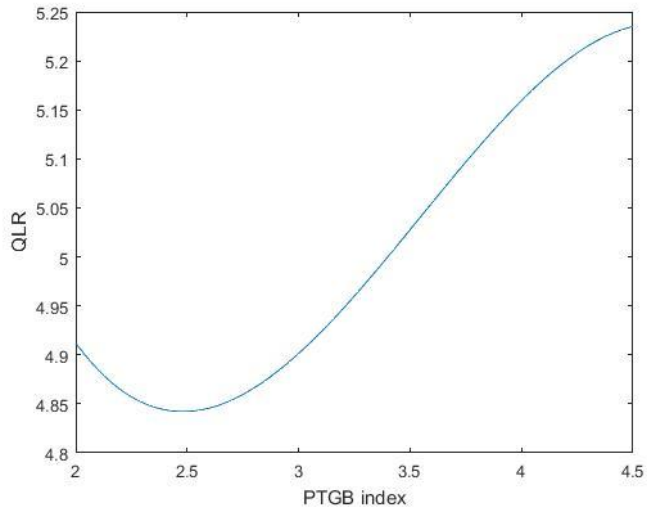

Figure 13 the relation between Happiness Index and PTGB

Second, analyze the relationship between plastic recycling rate and PTGB. It is found that the univariate nonlinear regression performs best. The analysis result is shown in the following equation. The function image is shown in Figure 14. With the promotion of plastic limit policies and measures, the plastics recycling rate has gradually increased, indicating that the plastic limit policies and measures are positive and effective.

$$EI = 0.65 \times z^2 + 4 \times z + 36, \quad where \ z = (PTGB - 3.4) / 0.73 \tag{16}$$

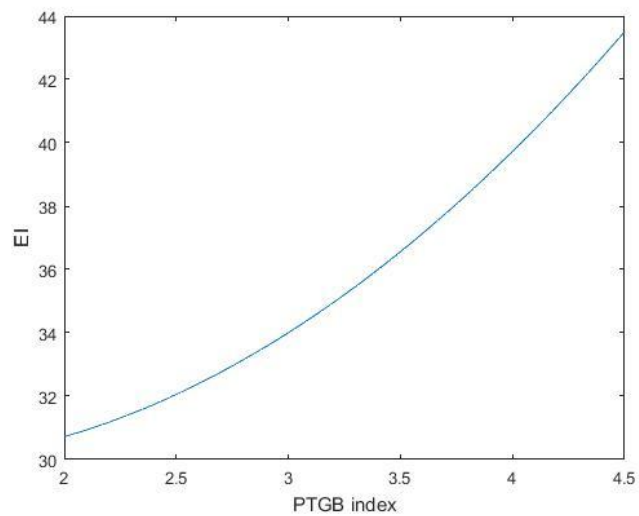

Figure 14 the relationship between plastic recycling rate and PTGB

At last, we analyze the relation between the annual growth rate of plastic production and PTGB using univariate non-linear regression which has the best fitting result. The analysis result is shown in the formula 17. The function image is shown in Figure 15. With the promotion of plastic limit policies and measures, the annual growth rate of plastic output has gradually decreased, indicating that plastic limit policies play a positive role in environment protection and restraint the development of plastic industry significantly.

$$IPID = -0.0055 \times PTGB + 0.057 \tag{17}$$

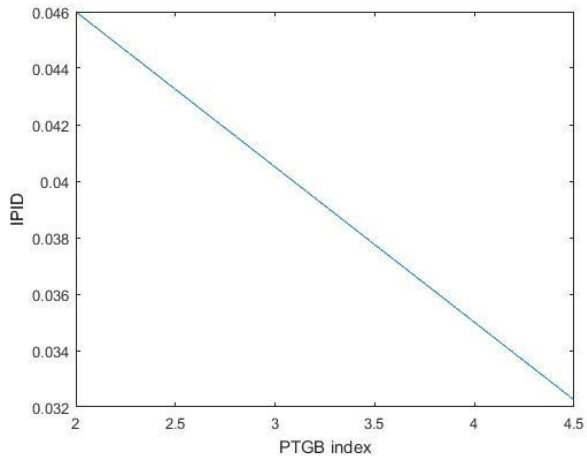

Figure 15 The relationship between the annual output growth rate of plastics and PTGB index

8. Distribution model based on carbon trading ideas and range transformation method

8.1 Basic analysis

For task 4, the disposable plastic waste is a global problem reflecting everyone but the causes and effects are not equally distributed across nations or regions. In order to solve the equity issues, different nations or regions shoulder different responsibilities. This paper studies how should different nations or regions do to maintain our environment fairly. We set a global disposable plastic usage limit based on the global environment capacity calculated in task1. Then the limit is assigned to every country or regions with a weight based on the fully consideration of their realistic situations (For example, economic, peace level and industrialization, etc.) Next, normalize the weight, multiply the normalized weight to global environment capacity to get the disposable plastic waste usage quota of each country or regions for the year. Now we are confident to say that each country or region has a fair share of disposable plastic waste usage quota. Besides, we are inspired from carbon trading models. The carbon trading model rules that when the country's usage exceeds the quota, the country needs to buy it from other countries' quota surplus; when the country has surpluses, the quota can be sold or not used.

By consulting the literature, this paper sets three positive indicators the density of population (PD), environmental performance index (EPI) and global peace index (GPI) , also sets three negative indicators GDP , industrial production (IP) and the final consumption expenditure per capita (C), where the value of positive indicators have the same trend with disposable plastic waste usage quota and the value of negative indicators is just the opposite.

8.2 Distribution model based on range transformation method

Firstly, we collect every country or region's data of six indicators in 2018. Then, using entropy weight method to get the weights of six indicators objectively. The six weights are:

$$H = \begin{pmatrix} 0.2222 & 0.2245 & 0.1434 & 0.1306 & 0.0608 & 0.2186 \end{pmatrix}$$

Secondly, collect m countries or regions' data of six indicators. Then we have matrix G with m rows and six columns as follows, where each column represents one indicator.

$$G = \begin{array}{c} \begin{array}{ccccc} PD & EPI & GPI & GDP & IP & C \end{array} \\ \begin{pmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{16} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{26} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{m,1} & x_{m,2} & x_{m,3} & \cdots & x_{m,6} \end{pmatrix} \end{array} \tag{18}$$

Next important step is the range transformation on G. For positive indicators, the transformation formula is shown in (19). For negative indicators, the transformation formula should be (20). After transformation, we get new matrix G' with six positive indicators.

$$y_{ij} = \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}}, (1 \le i \le m, 1 \le j \le 6) \tag{19}$$

$$y_{ij} = \frac{x_j^{\max} - x_{ij}}{x_j^{\max} - x_j^{\min}}, (1 \le i \le m, 1 \le j \le 6) \tag{20}$$

$$G' = \begin{pmatrix} y_{11} & y_{12} & y_{13} & \cdots & y_{16} \\ y_{21} & y_{22} & y_{23} & \cdots & y_{26} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_{m,1} & y_{m,2} & y_{m,3} & \cdots & y_{m,6} \end{pmatrix}$$

Finally, calculate $G' \times H$ to get matrix F, where each row represents the value of each country or region. Normalize F to get F'. F' times environmental capacity to obtain the quota of disposable plastics usage in each country or region.

$$F = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix} \qquad F' = \begin{pmatrix} \dfrac{v_1}{\sum_{i=1}^{m} v_i} \\ \dfrac{v_2}{\sum_{i=1}^{m} v_i} \\ \vdots \\ \dfrac{v_m}{\sum_{i=1}^{m} v_i} \end{pmatrix}$$

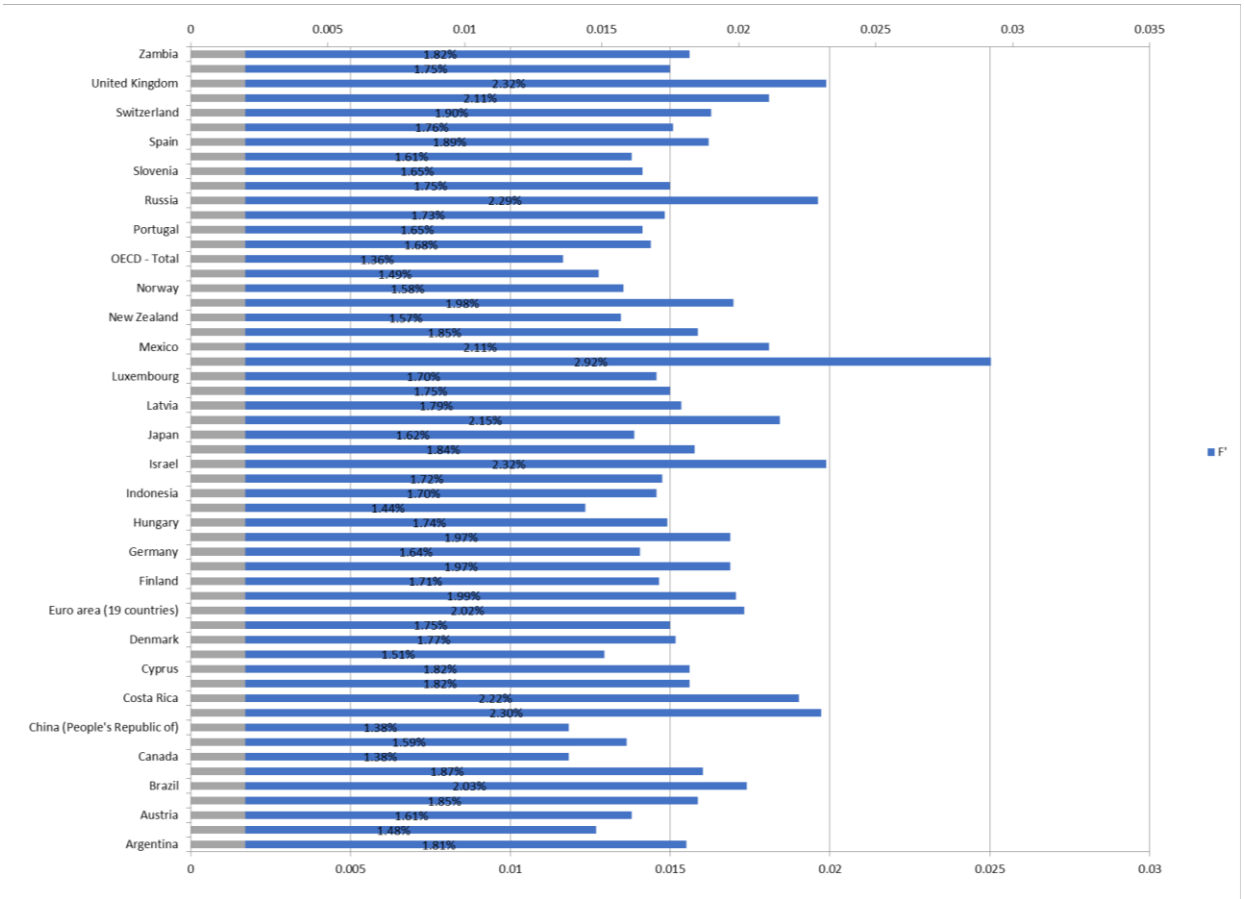

Figure 16 Percentage of national quotas

Since each country or region gains their annual disposable plastic usage quota, they should organize the consumption of the limited quota every year because no one want to accept penalties or be forced to spend large sum of money for other countries' quota surplus. It also encouraging countries or regions to protect our environment by selling the disposable plastic quota surplus. Overall, our distribution model not only fully considers the realistic situation of each country, but also builds principles of helping each other and to reducing our shared disposable plastic usage. It considers both fairness and environmental protection.

9. Sensitivity Analysis

There are estimation errors and measurement errors in statistical data which may affect the accuracy of model results. In this part, we analyze the sensitivity of the optimization model in task 2 to test the robustness of the model. We use LINGO to analyze the related factors of the maximum reduction in plastic waste. The operation results are demonstrated below.

As the figure shows, x, y, z, and w keeps the optimal solution in the range of 0 to INF.

Ranges in which the basis is unchanged:

Objective Coefficient Ranges

Variable	Current Coefficient	Allowable Increase	Allowable Decrease
X	0.6138000	INFINITY	0.6138000
Y	0.6167000	INFINITY	0.6167000
Z	0.6167000	INFINITY	0.6167000
W	0.5900000	INFINITY	0.5900000

Righthand Side Ranges

Row	Current RHS	Allowable Increase	Allowable Decrease
2	-936.9900	INFINITY	276.3504
3	-626.2871	INFINITY	208.2374
4	834.5245	INFINITY	208.2374
5	-213.0774	INFINITY	306.2790
6	519.3564	INFINITY	306.2790
7	-225.2129	INFINITY	161.5999
8	386.8128	INFINITY	161.5999
9	-76.62260	INFINITY	164.5210
10	241.1436	INFINITY	164.5210

From the above figure, y and w, as the banned disposable plastic usage of in the city and countryside respectively, have relatively high sensitivity with large change interval and coefficient. So, the changes in y and w can affect the results of the model to a certain extent. x and y, as the disposable plastic replacement in the city and countryside respectively, have relatively low relevance to the target. Also, when the number of fits increases, the residuals are decreasing and always in a 95% confidence interval. As a result, our optimization model is reliable.

10. Strengths and Weaknesses

- **Strengths**
- This paper introduces the latest data from many authoritative institutions (such as the EPS platform) to follow the actual situation of development in various countries. This guarantees the objectivity and rigor of the research source
- This paper considers a variety of indicator systems from different subjects (such as Economics, Environmental Science, Political Science, etc.), which shows the fully analysis of our environmental situations.
- **Weakness**
- The performance of sensitivity of optimization model in task 2 is not good enough to make our results very convincing.
- When collecting some sample data, due to limited channels limitation, some study resource only covers most countries in the world. It may affect the accuracy of distribution model in task 4.

Memo

From: Team # 2016845

To: ICM

Date: February 17,2020

Subject: The Timeline to Reach a Great Goal

Dear Sir,

Since the 1950s, the rapid development of the plastics industry has greatly facilitated human life but also brought about the urgent problem of disposable plastic waste. Today, many plastics are produced on the planet, and over half of them are disposable plastics. Most disposable plastics are non-degradable and discarded only on average 15 minutes usage. Then the massive disposable plastic waste is formed and piled up everywhere on the earth.

Write a two-page memo to the ICM describing a realistic global target minimum achievable level of global single-use or disposable plastic product waste,

Our team represent you with some recommendations how you might do to reduce the global disposable product waste to the environmental safe level in your next 20 years.

Firstly, we analyzed the environmental capacity of disposable plastic waste without further harming the environment. We studied that the environmental capacity of disposable plastic waste is equal to the sum of the harmless incineration, the disposable plastic waste recycling, and the natural self-purification. The relationship is shown in the figure below.

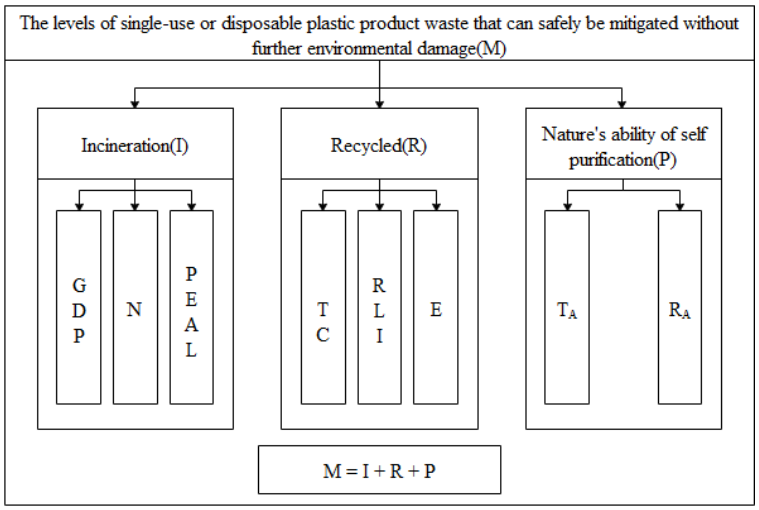

Figure 1

Secondly, we analyzed to what extent the disposable plastic waste can be reduced. The maximum reduction of disposable plastic waste is related to the source and use of plastic waste, residents' satisfaction, policies and measures, and geographical differences. In order to combine these factors, we build an optimization model based on the sigmoid function and simulated annealing algorithm to calculate the maximum reduction of disposable plastic waste.

Thirdly, we set an achievable goal of global disposable plastic waste level based on the study of relation between environment capacity and our goal. Its impact on the lives of residents,

the ecological environment, and the development of the plastic industry respectively is also fully analyzed in our optimization model.

For last thing, we put forward a solution for the global equity issues. Our team believes that the annual environmental capacity of disposable plastic waste is the total disposable plastic usage worldwide. Through some reasonable indicator's analysis and the inspiration from carbon trading rules, we have determined the weight for each country to obtain the corresponding proportion of annual disposable plastic usage.

By consulting the literature, it is impossible to reach the goal of global plastic waste in the second for now. But we are confident to achieve 40% of the goal by 2015, 70% of the goal by 2030, and 100% of the goal by 2040 considering the rapid development of technology, economy and civilization.

However, the condition discussed above is no irresistible case occurrence. When something unexpected happens, the timeline would change. Some can accelerate the achievement of goals (breakthrough technology development, soaring environmental awareness of residents, unprecedented strength of plastic pollution control policies, etc.), and others may obstruct human progress towards environmental protection. (natural disasters such as earthquakes, tsunamis, volcanic eruptions, mountain fires etc. virus invasion such as China's new coronavirus, SARS, Ebola, etc.)

We hope our analysis of disposable plastic waste problem may contribute to your recognition of environment situations. We welcome additional inquiries about the disposable plastic waste.

With Best Regards.

Agent team

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Appendices

Here are some codes and data

year	I	GDP	N	R	TC	RLI	P
2017	1163.70375	832035.9	286	1380.809	9538.95	12637.685	142
2016	1014.53	746395.1	249	1314.764	9219.8	11487.43	119
2015	849.13125	688858.2	220	1281.081	8806.3	10962.49	99
2014	732.86125	643563.1	188	1293.159	9575.5	10672.67	82.4
2013	637.13375	592963.2	166	1261.503	9037.2	9888.195	67.5
2012	492.81375	538580	138	1130.171	8253.46	8865.78	54
2011	357.40375	487940.2	109	1025.988	7114.03	7917.71	45
2010	318.54625	412119.3	104	1032.017	7612.19	6270.65	34.6
2009	278.025	348517.7	93	939.5072	5258.39	4835	27

year	PEAL	TC	GDP	BPP	HI	DPRR	RP
2017	0.107	9538.95	832035.9	730.2	5.25	41.9	3.30%
2016	0.1049	9219.8	746395.1	650.7	5.27	42.4	3.42%
2015	0.1028	8806.3	688858.2	510.8	5.25	39.9	3.54%
2014	0.1008	9575.5	643563.1	443.2	5.14	38.9	3.67%
2013	0.0989	9037.2	592963.2	402.1	4.98	36.6	3.81%
2012	0.0969	8253.46	538580	382.2	4.98	35	3.96%
2011	0.095	7114.03	487940.2	356.1	4.95	33.7	4.12%
2010	0.0932	7612.19	412119.3	321.3	4.93	32.5	4.30%
2009	0.0913	5258.39	348517.7	289.7	4.91	31.9	4.49%

```
clear
close all
clc

%The introduction of data
A=xlsread('data3.xls');

%year
year=A(:,1);

%The proportion of cities
bl=A(:,2)./(A(:,2)+A(:,3));

%% Fit two functions
%Substitution function
```

```
%plot(year,A(:,6).*bl,'r*')
%Forbidden function
%plot(year,A(:,7).*bl,'r*')

%% Solve for the value range
%Alternative range
x=year;
z=(x-2*10^3*eye(size(A(:,1))))/3;
y=-58*z.^2+2.1*(10^2)*z+3.1*10^3*eye(size(A(:,1)));
max(y);
min(y);
%Forbidden range
x=2017;
y=0.36*x.^2-1.4*10^3*x+1.4*10^6*eye(size(x));
max(y);
min(y);

%% Solve for two coefficients
%Coefficient 1: degree of substitution
%The normalized
x1=A(:,11).*bl
y1=(x1-min(x1).*ones(size(x1)))./(max(x1)-min(x1));
x2=A(:,12).*bl;
y2=(x2-min(x2).*ones(size(x2)))./(max(x2)-min(x2));
x3=A(:,13).*bl;
y3=(x3-min(x3).*ones(size(x3)))./(max(x3)-min(x3));
%The weighted
y0=0.4*y1+0.2*y2+0.4*y3
%Sigmod function
y00=ones(size(y0));
for i=1:11;
   y00(i)=1/(1+exp(1)^(-y0(i)))
end
```